Justification-Quality Rubric

for evaluating textual justifications produced by
generative-AI source-code similarity analysts

Applied to the 480 responses (4 LLMs $\times$ 120 cases) of the SimilCode benchmark

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1 Purpose and scope

This document formalises the rubric that was applied by the principal investigator to rate the textual justifications produced by the four commercial large language models Claude Opus 4.1, GPT-5, Gemini 2.5 Pro, and DeepSeek-V3 when queried as source-code similarity analysts in the within-subjects benchmark reported in the accompanying manuscript. The rubric was applied to each of the 480 model responses in the benchmark (4 models × 120 paired cases).

The rubric operationalises three dimensions of justification quality: **clarity**, **completeness** and **coherence**. These three dimensions and their associated indicators reproduce verbatim the specification recorded in the data-collection template of the underlying degree-thesis project (Annex 6 of the source-code similarity study conducted at Universidad Técnica Estatal de Quevedo).

2 Design principles

The rubric was designed to satisfy three requirements:

1. **Traceability to the study prompt.** The completeness dimension covers exactly the four similarity sub-dimensions the models were required to justify in the standardised prompt: lexical, structural, functional, and stylistic (see Appendix A of the manuscript).
2. **Consistency with the Chain-of-Verification block.** The coherence dimension operationalises the two verification conditions the models were instructed to satisfy: consistency between the numerical score and its textual justification, and internal logical consistency of the response.
3. **Reproducibility across evaluators.** All indicators are stated as observable properties of the model output, so that any evaluator with access to the response can apply the rubric independently.

3 Structure of the rubric

The rubric returns **nine indicators** organised across the **three dimensions** summarised in Table 1.

Table 1: Overview of the three rubric dimensions.

Dimension	Focus	Scale	Items
Clarity	Comprehensibility of the explanation, correctness of technical terminology, and logical structure of the response.	Ordinal 1–5	3
Completeness	Coverage of the four required similarity sub-dimensions in the justification.	Nominal Yes/No	4
Coherence	Consistency between numerical scores and textual justifications, and internal logical consistency.	Ordinal 1–5	2

4 Indicators

4.1 Dimension 1 — Clarity (ordinal 1–5)

#	Indicator	What the evaluator inspects
C1	Comprehensibility of the explanation.	Whether a reader with basic programming knowledge can follow the justification without additional context.
C2	Correct use of technical terminology.	Whether the technical vocabulary used by the model is applied consistently with its standard meaning in the software-engineering domain.
C3	Logical structure of the response.	Whether the justification follows an ordered exposition (premise → evidence → conclusion) rather than an unstructured enumeration.

Each indicator is scored on a 5-point ordinal scale: **1** = deficient; **2** = limited; **3** = acceptable; **4** = good; **5** = excellent.

4.2 Dimension 2 — Completeness (nominal Yes/No)

#	Indicator	What the evaluator inspects
P1	Coverage of lexical similarity.	Whether the justification explicitly discusses the lexical similarity score.
P2	Coverage of structural similarity.	Whether the justification explicitly discusses the structural similarity score.
P3	Coverage of functional similarity.	Whether the justification explicitly discusses the functional similarity score.
P4	Coverage of stylistic similarity.	Whether the justification explicitly discusses the stylistic similarity score.

Each indicator is scored as **Yes** (present) or **No** (absent).

4.3 Dimension 3 — Coherence (ordinal 1–5)

#	Indicator	What the evaluator inspects
H1	Consistency between numerical score and textual justification.	Whether the value of the numerical score matches the polarity and strength of the corresponding textual justification (a high score should not be justified with predominantly negative statements, and vice versa).

#	Indicator	What the evaluator inspects
H2	Internal logical consistency of the explanation.	Whether the justification is free of internal contradictions between the individual dimension scores and the overall similarity score.

Each indicator is scored on a 5-point ordinal scale: **1** = deficient; **2** = limited; **3** = acceptable; **4** = good; **5** = excellent.

5 Scoring sheet

Table 5 reproduces the scoring sheet applied to each of the 480 model responses. One sheet is completed per (*case*, *model*) combination.

Table 5: Scoring sheet applied per (case, model) combination.

Indicator	**Claude Opus 4.1**	**GPT-5**	**Gemini 2.5 Pro**	**DeepSeek-V3**
Clarity (1–5)				
C1 — Comprehensibility of the explanation C2 — Correct use of technical terminology C3 — Logical structure of the response				
Completeness (Yes/No)				
P1 — Coverage of lexical similarity P2 — Coverage of structural similarity P3 — Coverage of functional similarity P4 — Coverage of stylistic similarity				
Coherence (1–5)				
H1 — Score–justification consistency H2 — Internal logical consistency of the response				

6 Aggregation

From the raw indicator scores, the following aggregate quantities are computed for each (*case*, *model*) combination:

- **Clarity score:** arithmetic mean of C1, C2, C3.
- **Completeness score:** number of Yes marks among P1–P4 (integer 0–4).
- **Coherence score:** arithmetic mean of H1, H2.

- **Overall justification-quality score:** arithmetic mean of the three dimension scores after linearly re-scaling the completeness score from $[0, 4]$ to $[1, 5]$.

The overall justification-quality score is the dependent variable analysed by the Kruskal–Wallis test reported in the manuscript.

7 Citation

If you use this rubric, please cite:

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This rubric is derived directly from the data-collection template (Annex 6) of the source degree-thesis project that produced the SimilCode benchmark. The three dimensions (clarity, completeness, coherence), the nine indicators, and the associated scales are transcribed from that source. Descriptive labels of the ordinal scale (deficient, limited, acceptable, good, excellent) follow the conventional labelling of a five-point Likert scale and are not derived from a separate source in the project.